Machine Learning

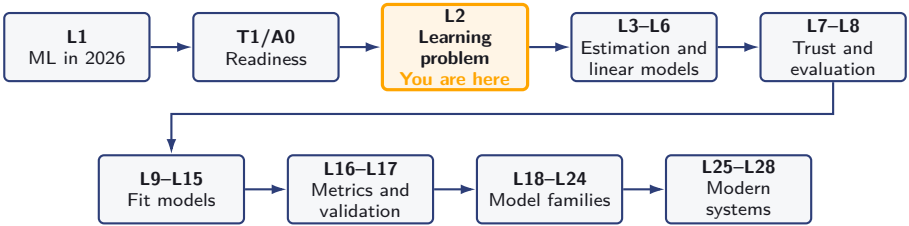
Lecture 2: Mathematical Foundations for Learning from Data

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Course Roadmap: You Are Here



Lecture 2 gives the mathematical language needed to move carefully between one observation, a dataset, a prediction rule, and an average loss.

Recall: Words We Used Last Time

Term	Working meaning
conditional probability	$p(y \mid x)$: what we believe about Y after observing $X = x$
total probability	$p(y) = \sum_x p(y \mid x)p(x)$, or an integral when X is continuous
non-stationarity	the relationship or distribution changes: $\mathcal{D}_{\text{past}} \neq \mathcal{D}_{\text{future}}$
weights $\boldsymbol{w}$	adjustable numbers that determine a prediction, e.g. $\hat{y} = \boldsymbol{w}^\top \boldsymbol{x} + b$
optimization	choosing parameters that make an objective as small as possible
convexity	a shape with no higher "valley" between two points; local minima are global minima

A Better Answer To The ChatGPT Question

Question

Why can a language model often help with substantial software, yet sometimes give an unconvincing answer to a very short numerical series?

At its core, a language model estimates likely continuations:

$$p_\theta(\text{next token} \mid \text{preceding text}).$$

- **Programming:** prompts often name a familiar task family; code patterns are abundant; execution and tests can reject a proposed answer.
- **Short series:** many rules g can fit the same few values; the examiner's intended rule may not be specified by the prefix alone.
- **Tools matter:** executing code or checking a symbolic rule gives feedback that plain next-token prediction does not automatically provide.

Lecture Objectives

By the end of this lecture, you should be able to:

1. state whether a symbol is a scalar, vector, matrix, function, or distribution, and check its dimension;
2. formalize a supervised learning problem from one example through a fitted function;
3. explain the four choices hidden inside a learning rule and what $\arg\min$ returns; and
4. distinguish discriminative and generative modelling targets, then move from one prediction error to empirical risk.

One Learning Example

For one observed event, let

$$\boldsymbol{x}_i = \begin{bmatrix} \text{rainfall} \\ \text{catchment area} \\ \text{slope} \end{bmatrix} \in \mathbb{R}^3, \qquad y_i = \text{observed flood depth} \in \mathbb{R}.$$

We choose a prediction rule f_θ and compute

$$\hat{y}_i = f_\theta(\boldsymbol{x}_i), \qquad \text{for a linear rule:} \quad \hat{y}_i = \boldsymbol{w}^\top \boldsymbol{x}_i + b.$$

Insight

The weights $\boldsymbol{w}$ and intercept b are parameters. Together, write $\theta = (\boldsymbol{w}, b)$. Learning means choosing their values from data.

What Are We Trying To Estimate?

Imagine an unknown data-generating relationship:

$$Y = f^\star(\boldsymbol{X}) + \varepsilon,$$

where $f^\star$ is the systematic relationship and ε represents noise, unmeasured causes, or measurement error.
We do not observe $f^\star$. We observe a finite sample

$$\mathcal{S}_{\text{train}} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^n$$

and construct an estimate $\hat{f}$ that should predict a future Y from a future $\boldsymbol{X}$.

Future-performance target

$$R(f) = \mathbb{E}_{(\boldsymbol{X},Y)\sim\mathcal{D}}\left[\ell\big(f(\boldsymbol{X}), Y\big)\right].$$

Machine Learning Is Four Explicit Choices

Choice	Notation	Question it answers
data	$\mathcal{S}_{\text{train}}$	Which examples are allowed to teach us?
model class	$\mathcal{F}$	Which functions f could we fit?
loss	$\ell(\hat{y}, y)$	Which prediction mistakes matter, and by how much?
algorithm	$\mathcal{A}$	How do we search among candidates in $\mathcal{F}$?

$$\mathcal{A} : \mathcal{S}_{\text{train}} \longmapsto \hat{f} \in \mathcal{F}.$$

Warning

There is no model without these choices, even when a software library hides them.

What Does arg min Mean?

Let $L(\theta)$ be an objective that assigns one number to each candidate parameter value.

$\min_{\theta} L(\theta)$ is the smallest objective value, $\arg\min_{\theta} L(\theta)$ is the parameter achieving it.

For the toy function $L(u) = (u-3)^2$,

$$\min_{u} L(u) = 0, \qquad \arg\min_{u} L(u) = 3.$$

In supervised learning, empirical-risk minimization is

$$\hat{f} \in \arg\min_{f \in \mathcal{F}} \underbrace{\frac{1}{n}\sum_{i=1}^{n} \ell\big(f(\boldsymbol{x}_i), y_i\big)}_{\widehat{R}_{\mathcal{S}_{\text{train}}}(f) \text{ computed from observed data}}.$$

Convexity Is Always With Respect To A Variable

A scalar function g is convex on a set if, for every u, v in the set and $0 \le \lambda \le 1$,

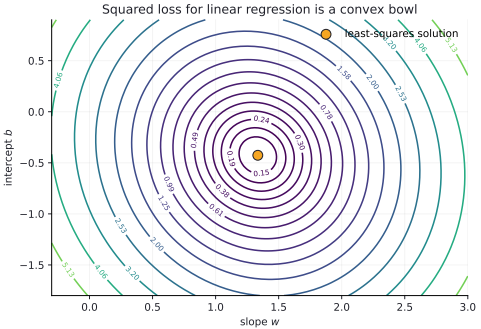
$$g\big(\lambda u + (1-\lambda)v\big) \le \lambda g(u) + (1-\lambda) g(v).$$

The question must name the variable

For fitting, the relevant object is the objective $L(\theta)$: is it convex in the *parameters* θ? A prediction rule can depend nonlinearly on the input $\boldsymbol{x}$ while its training objective is convex in θ.

$$L(\boldsymbol{w}, b) = \frac{1}{n} \sum_{i=1}^{n} \big(\boldsymbol{w}^\top \boldsymbol{x}_i + b - y_i\big)^2 \quad \text{is convex in } (\boldsymbol{w}, b).$$

Seeing The Objective: A Convex Bowl



Insight

Each contour collects parameter pairs (w, b) with equal training loss. The marked point is the arg min: a *location* in parameter space, not a loss value.

Mathematical Objects Have Types

The notation is not decoration. Each symbol names a different kind of object.

Object	Typical symbol	Meaning
scalar	$y_i,\ \hat{y}_i \in \mathbb{R}$	one observed or predicted number
vector	$\boldsymbol{x}_i \in \mathbb{R}^d$	the d features for one observation
matrix	$\boldsymbol{X} \in \mathbb{R}^{n \times d}$	n observations, each with d features
function	$f_\theta : \mathcal{X} \to \mathcal{Y}$	a rule with adjustable parameters θ
distribution	$\mathcal{D}$	the unknown process that can generate observations

Insight

When two expressions are confused, first ask: are they even the same type of object?

Dimensions Are A Correctness Check

For a linear prediction rule, let

$$\boldsymbol{x}_i \in \mathbb{R}^d, \qquad \boldsymbol{\theta} \in \mathbb{R}^d, \qquad \boldsymbol{X} \in \mathbb{R}^{n \times d}.$$

Then one prediction and all predictions have compatible dimensions:

$$\underbrace{\boldsymbol{x}_i^\top}_{1\times d}\underbrace{\boldsymbol{\theta}}_{d\times 1} = \underbrace{\hat{y}_i}_{1\times 1}, \qquad \underbrace{\boldsymbol{X}}_{n\times d}\underbrace{\boldsymbol{\theta}}_{d\times 1} = \underbrace{\widehat{\boldsymbol{y}}}_{n\times 1}.$$

Warning

If the inner dimensions do not agree, the expression is not merely inconvenient: it is undefined.

Two Mathematical Views Of Prediction

Discriminative

Model the conditional relationship

$$p_\theta(y \mid \boldsymbol{x})$$

or directly learn $f_\theta(\boldsymbol{x})$. For classification:

$$\hat{y} = \arg\max_y p_\theta(y \mid \boldsymbol{x}).$$

Generative

Model how inputs and outputs arise jointly:

$$p_\theta(\boldsymbol{x}, y) = p_\theta(\boldsymbol{x} \mid y) p_\theta(y).$$

Then derive

$$p_\theta(y \mid \boldsymbol{x}) = \frac{p_\theta(\boldsymbol{x} \mid y) p_\theta(y)}{p_\theta(\boldsymbol{x})}.$$

Insight

Both can predict. Generative models additionally support questions about how data could have been produced, sampled, or made missing.

One Question, Asked Two Ways

It rained **90 mm** today.

Will there be a flood?

One input: $x =$ daily rainfall in mm.

One label: $y \in \{\text{flood},\ \text{no flood}\}$.

There are two honest ways to answer. They differ in *how much of the world you bother to model*.

Route 1 — Model How Rainfall Behaves

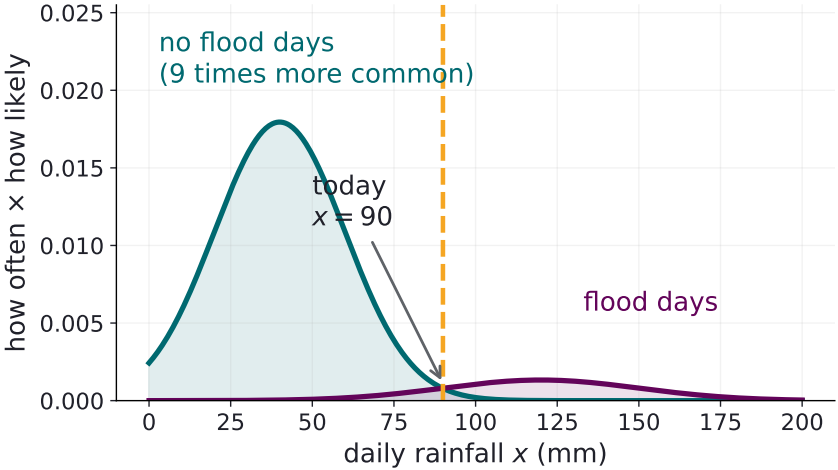
Describe what a rainy day looks like *separately* for flood days and for dry days, and say how often floods happen at all.

	how often	rainfall on such a day
flood days	0.10	$\mathcal{N}(120, 30^2)$
dry days	0.90	$\mathcal{N}(40, 20^2)$

Insight

This is a story about how the data *arose*. It is enough to invent a plausible rainfall figure for tomorrow — which is more than we need to answer the question.

Route 1, Pictured



Each curve is already weighted by how often that day happens. Today lands where they cross.

Route 1 — Turn The Story Into An Answer

Reading the two curves at $x = 90$ gives

$$p(x \mid \text{flood}) = 0.00807, \qquad p(x \mid \text{no flood}) = 0.000876.$$

In Class

Weight each by how often that kind of day happens, then normalise. Three lines.

Route 1 — Work It Out

Weight by how often, then normalise.



Route 2 — Model Only The Cut

Never ask how rainfall is distributed. Go straight at the thing we were asked for:

$$p_\theta(\text{flood} \mid x) = \sigma(wx + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Insight

No $p(x)$ appears anywhere. Two numbers place one cut on the rainfall axis. This model cannot invent tomorrow's rainfall — and was never asked to.

In Class

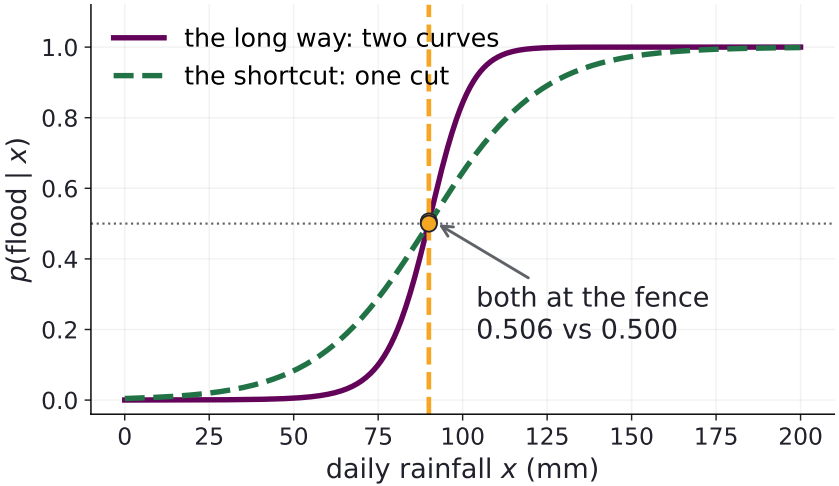
Fitting on past data gave $w = 0.06$ and $b = -5.4$. Evaluate it at today's $x = 90$.

Route 2 — Work It Out

Evaluate $\sigma(wx + b)$ at $x = 90$, with $w = 0.06$ and $b = -5.4$.



Both Routes, One Picture



Different shapes, same crossing. They disagree away from the boundary, not at it.

Same Decision, Different Confidence

x (mm)	60	90	120
Route 1, generative $p(\text{flood} \mid x)$	0.016	0.506	0.996
Route 2, discriminative $\sigma(wx + b)$	0.142	0.500	0.858

Both put today at the fence, and both call 60 mm dry and 120 mm wet. They disagree only in how confident they are away from the boundary.

Insight

The generative model spent five numbers (two means, two spreads, one prior) learning how rainfall behaves; the discriminative model spent two, learning only where to cut. Same decision, different amount of story — and only the first can simulate a plausible rainfall day or cope with x going missing.

The Statistical Setup

Data-Generating Distribution

Let

$$Z = (\boldsymbol{X}, Y) \sim \mathcal{D},$$

where $\mathcal{D}$ is the unknown distribution over inputs and outputs.
The training set is often idealized as

$$\mathcal{S}_{\text{train}} = \{Z_1, \ldots, Z_n\}, \qquad Z_i \overset{\text{iid}}{\sim} \mathcal{D}.$$

The iid assumption means examples are independent and drawn from the same distribution.

Why The iid Assumption Is A Model

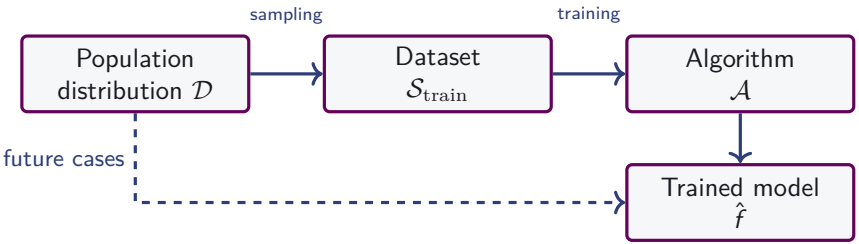
The iid assumption can be false in many real datasets:

- time series: tomorrow depends on today
- spatial data: nearby locations are correlated
- repeated users or assets: multiple records are not independent
- policy changes: data before and after intervention differ
- sensors: missingness and measurement error have structure

Warning

Many ML mistakes begin when a dataset is treated as iid because the spreadsheet rows look independent.

Three Objects To Keep Separate



Summary

Do not confuse the world, the dataset, the algorithm, and the fitted model.

A Loss Function Scores One Prediction

Loss

A loss function

$$\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}_{\geq 0}$$

assigns a penalty to a prediction and a target.

Examples:

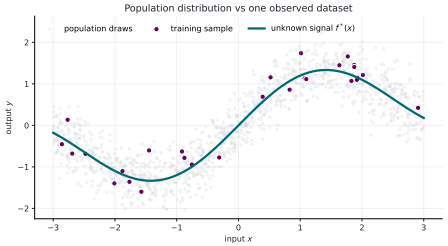
$$\ell_{\mathrm{sq}}(\hat{y}, y) = (\hat{y} - y)^2, \qquad \ell_{\mathrm{abs}}(\hat{y}, y) = |\hat{y} - y|.$$

$$\ell_{0/1}(\hat{y}, y) = \mathbb{1}\{\hat{y} \neq y\}.$$

Risk: The Quantity We Actually Care About

For a model f, the population risk is

$$R(f) = \mathbb{E}_{(\boldsymbol{X},Y)\sim\mathcal{D}}\left[\ell(f(\boldsymbol{X}), Y)\right].$$



Warning

Risk is about future draws from $\mathcal{D}$ — and we cannot compute it exactly, because we only ever hold one sample, never the whole grey cloud.

Empirical Risk: The Quantity We Can Compute

Given a dataset $\mathcal{S}_{\text{train}} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n}$,

$$\widehat{R}_{\mathcal{S}_{\text{train}}}(f) = \frac{1}{n} \sum_{i=1}^{n} \ell(f(\boldsymbol{x}_i), y_i).$$

Empirical Risk Minimization chooses

$$\hat{f} = \arg\min_{f \in \mathcal{H}} \widehat{R}_{\mathcal{S}_{\text{train}}}(f).$$

Insight

ERM is the workhorse idea behind much of supervised ML.

A Concrete ERM Example

For linear regression, $f_{\boldsymbol{w},b}(\boldsymbol{x}) = \boldsymbol{w}^\top \boldsymbol{x} + b$.
Training then chooses the parameter pair:

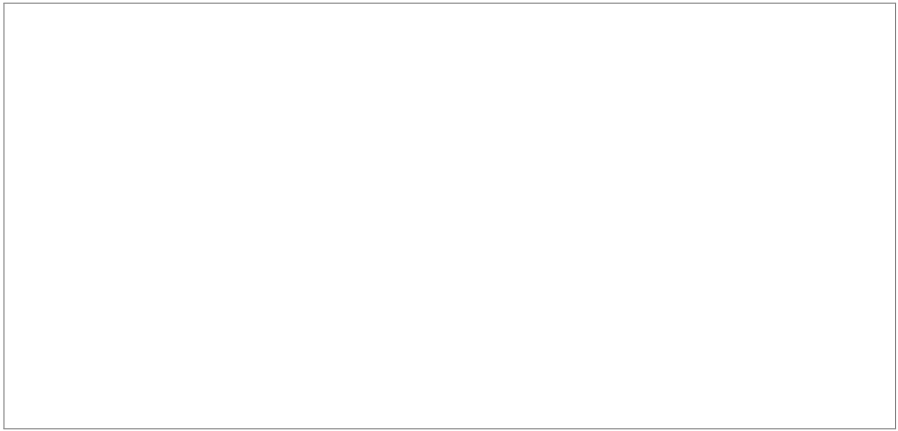
$$(\hat{\boldsymbol{w}}, \hat{b}) = \arg\min_{\boldsymbol{w},b} \widehat{R}_{\mathcal{S}_{\text{train}}}(\boldsymbol{w}, b).$$

In Class

Supply, in order: the residual r_i; the one-example squared loss ℓ_i; then the average over all n.

A Concrete ERM Example: Work It Out

Residual r_i, then one-example squared loss ℓ_i, then the average over all n examples.



Gaussian Noise Gives A Conditional Density

Assume the observed target differs from the systematic prediction by Gaussian noise:

$$Y = f_\theta(\boldsymbol{X}) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2).$$

$$Y \mid \boldsymbol{X} = \boldsymbol{x}, \theta \sim \mathcal{N}\big(f_\theta(\boldsymbol{x}), \sigma^2\big).$$

Therefore, for one observed pair $(\boldsymbol{x}_i, y_i)$,

$$p_\theta(y_i \mid \boldsymbol{x}_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\big(y_i - f_\theta(\boldsymbol{x}_i)\big)^2}{2\sigma^2}\right].$$

Insight

The squared residual has appeared because of the Gaussian noise assumption, not because someone simply preferred a square.

Derivation: Likelihood To Least Squares

If the n examples are conditionally independent given θ, the likelihood is a product, and logs turn it into a sum:

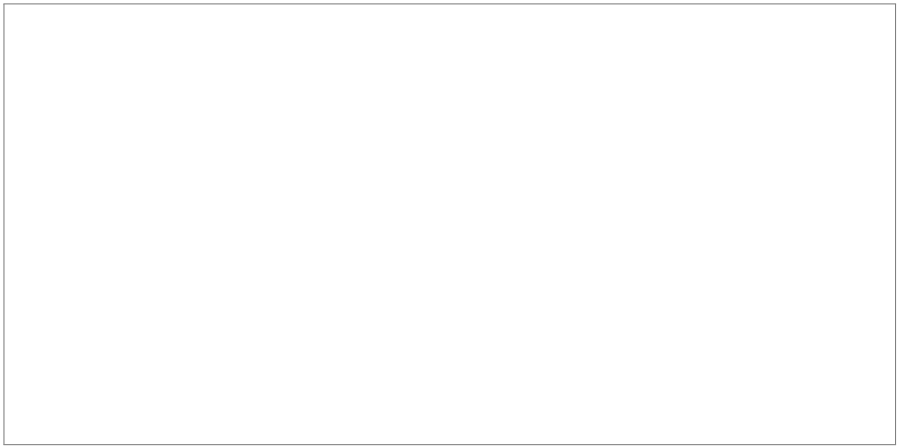
$$L(\theta) = \prod_{i=1}^{n} p_\theta(y_i \mid \boldsymbol{x}_i), \qquad \log L(\theta) = \sum_{i=1}^{n} \log p_\theta(y_i \mid \boldsymbol{x}_i).$$

In Class

Take the log of the single-example density from the previous slide. Which two terms appear, and which depends on θ?

Likelihood To Least Squares: Work It Out

Expand $\log p_\theta(y_i \mid \boldsymbol{x}_i)$. Mark the term that depends on θ.



From Log-Likelihood To Least Squares

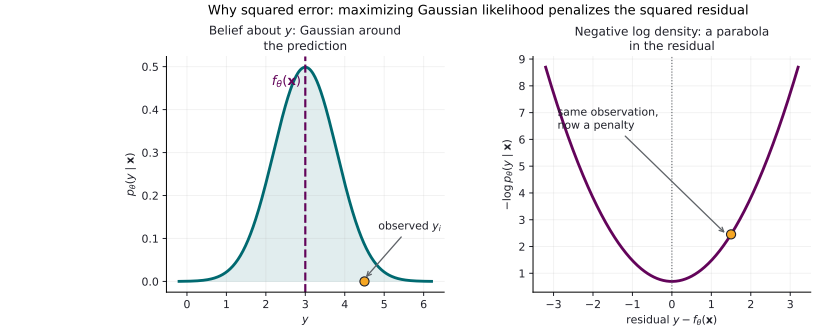
Summing the single-example result over the n examples,

$$\log L(\theta) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\big(y_i - f_\theta(\boldsymbol{x}_i)\big)^2.$$

The first term and the positive factor $1/(2\sigma^2)$ do not change the optimizing θ. Hence

$$\arg\max_\theta L(\theta) = \arg\max_\theta \log L(\theta) = \arg\min_\theta \sum_{i=1}^{n}\big(y_i - f_\theta(\boldsymbol{x}_i)\big)^2.$$

The Picture Behind The Algebra



Insight

The same observed y_i appears twice: once as a point under the Gaussian belief, once as a squared-residual penalty. Maximizing likelihood and minimizing squared error are one decision seen from two sides.

Worked Example: One Error To Empirical Risk

Suppose a model predicts flood depths for three observed events:

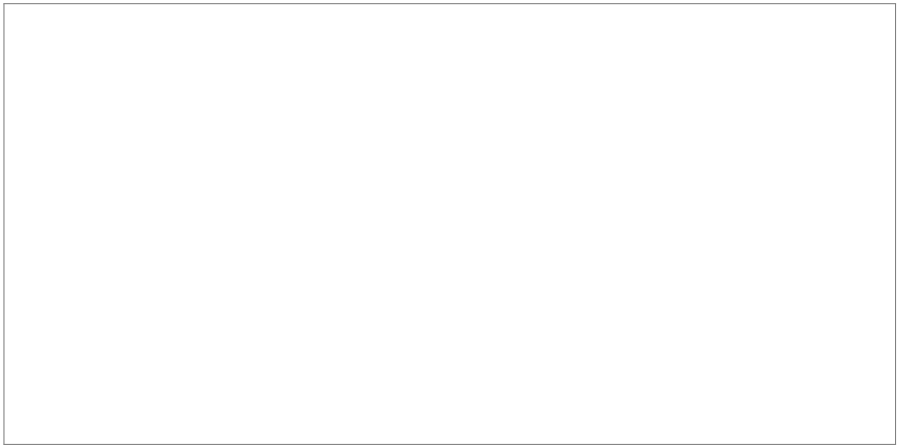
$$\boldsymbol{y} = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}, \qquad \widehat{\boldsymbol{y}} = \begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix}.$$

In Class

Work these out with me: the residual vector $\boldsymbol{r}$, then the empirical risk $\widehat{R}_{\mathcal{S}_{\text{train}}}(f)$ under squared-error loss. Keep the signs.

One Error To Empirical Risk: Work It Out

Residual vector $\boldsymbol{r} = \boldsymbol{y} - \widehat{\boldsymbol{y}}$, then $\widehat{R}_{\mathcal{S}_{\text{train}}}(f)$ under squared-error loss.



The Mathematical Spine

The entire first part of the course will reuse this chain:

$$\underbrace{\boldsymbol{x}_i}_{\text{one input}} \xrightarrow{f_\theta} \underbrace{\hat{y}_i}_{\text{one prediction}} \xrightarrow{\ell(\hat{y}_i, y_i)} \underbrace{\ell_i}_{\text{one scalar loss}} \xrightarrow{\frac{1}{n}\sum_i} \underbrace{\widehat{R}_{\mathcal{S}_{\text{train}}}(f_\theta)}_{\text{dataset-level objective}} .$$

Summary

Next lecture: turn this notation into data vectors, a linear prediction rule, and a least-squares objective that we can inspect geometrically.